Programming for AI

Usman Nazir

Introduction

Highlights of my Education and Experience

2012
Graduation
Analysis of energy efficient hierarchical routing protocols for WSNs
COMSATS

2013-2014
Research Assistant, SAHSE AdCom Lab, LUMS

2016
Post Graduation
Cryptographic Security for Traffic Engineering in SDX
LUMS

2016-2017
Lecturer, Statistics and Computer Sciences Department,
University of Veterinary and Animal Sciences

2017
Split-site PhD CS
Learning Socio-economic Indicators from Remote Sensing Data
LUMS

2022
Commonwealth scholarship
UNIVERSITY OF LEEDS
Split-site PhD CS
Learning Socio-economic Indicators from Remote Sensing Data using Graph Neural Networks

QGIS Lab Demonstrator,
School of Geography,
University of Leeds
UNIVERSITY OF LEEDS

Research Scientist [CITY at LUMS & University of Oxford, UK]
Aug 2022 – till now

Founder of PAAI (Pakistan Academy of Artificial Intelligence)

Assistant Professor at SCIT, BNU

★ publications

paai.tech

Center for AI Research (CAIR)

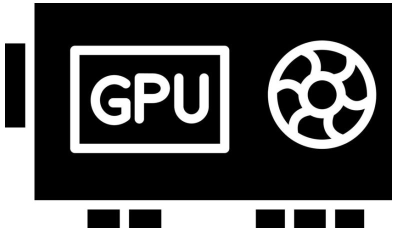
Usman Nazir
Assistant Professor
Group Head CAIR

CENTER FOR AI RESEARCH

ai.bnu.edu.pk

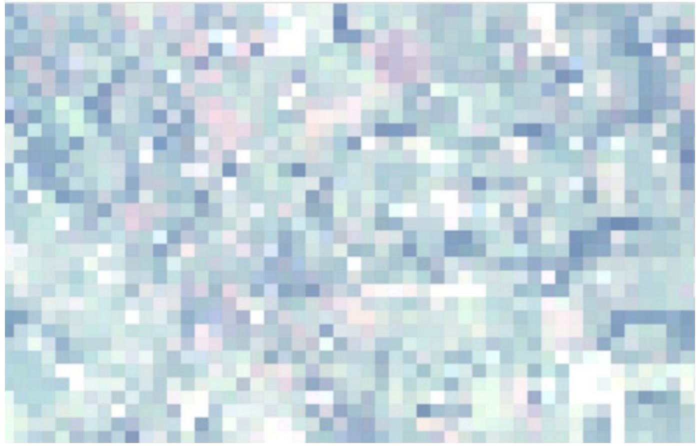
CAIR Infrastructure

State-of-the-art Infrastructure



10 Nvidia RTX 4070 GPU
1 Nvidia RTX 3080 GPU

Access to high-resolution satellite imagery



30 meter per Pixel

10 meter per Pixel

BNU Research

Research Publications and Impact

AI Semester Projects:

2 ICONIP 2024 Publications

1 AAAI 2024 Publication

2 NeurIPS publications (submitted)

A collaboration between the University of Oxford and Beaconhouse National University has combined the power of artificial intelligence with remote satellite imagery to help tackle the problem of unregulated brick kilns across South Asia. These kilns are a major source of air pollution, a key contributor to climate change, and notorious for people trafficking and modern-day slavery.

Research Threads

[Geo-AI]
Geospatial Analysis and Environmental Monitoring

[Vision-AI]
Computer Vision and Image Processing

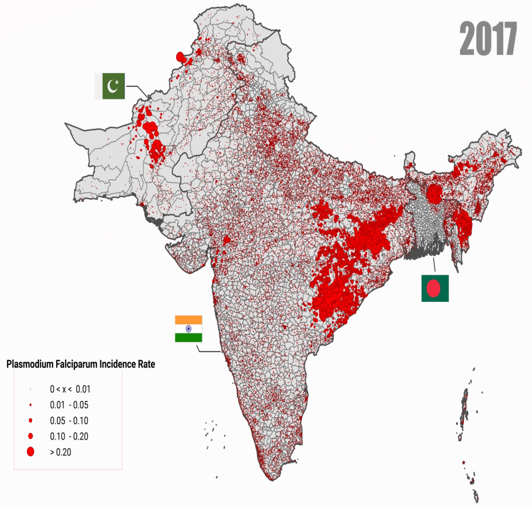
[Predict-AI]
Predictive Analysis and Behavioral Analysis

Geo-AI

LULC classification through remote sensing

Mapping Lahore's land use land cover -
Tree cover estimation is illustrated

Predicting Global Burden of Disease using Earth Observation

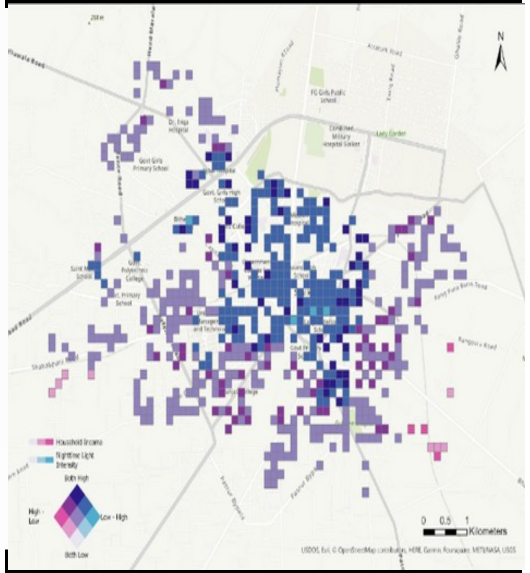


Malaria Incidence Rate [data collected Demographic Health Survey (DHS)] in South Asia

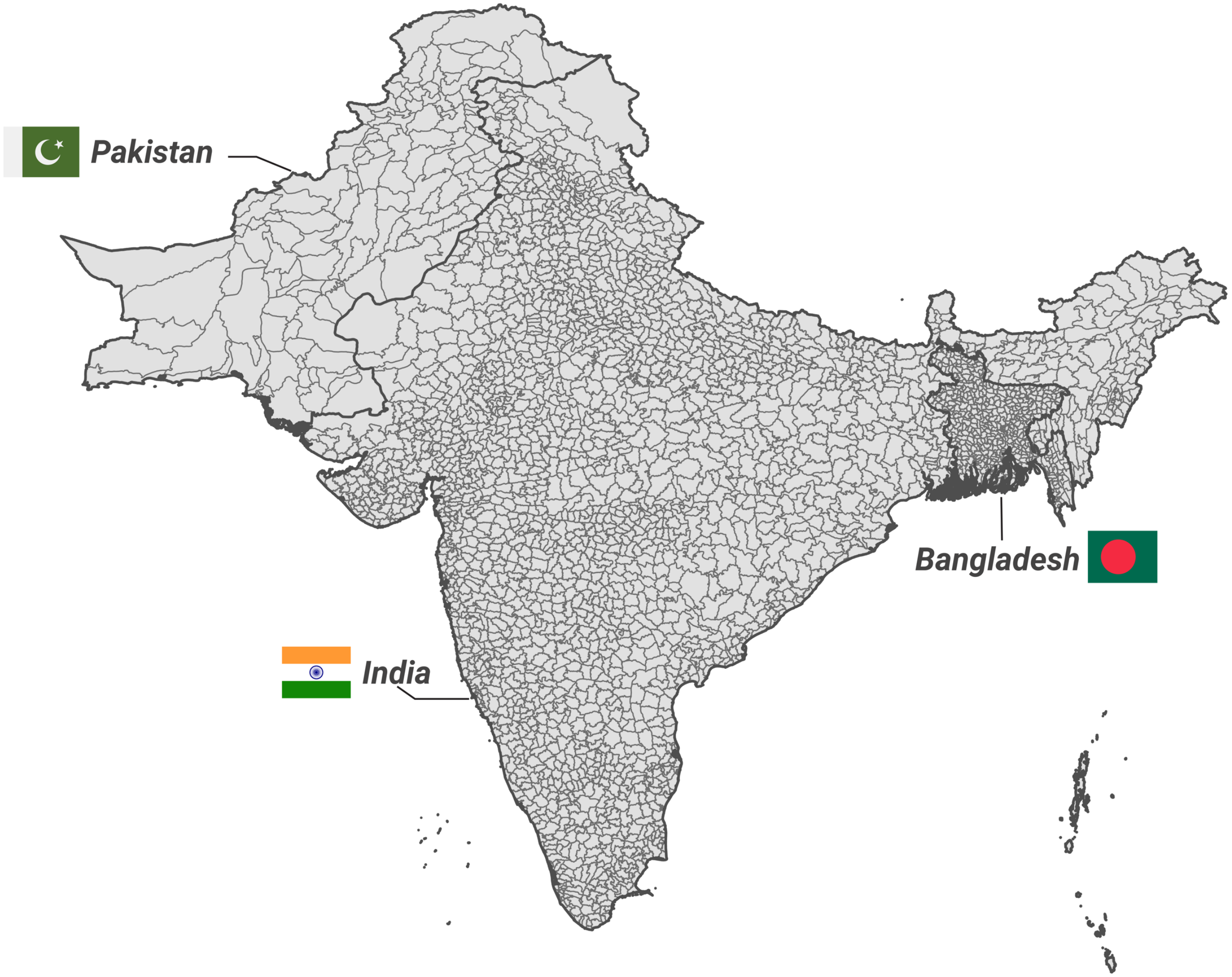
Waste Dumping in Oceans

Per Annum Water Quality Index and Chlorophyll Index for the Manchhar Lake (Largest freshwater lake in Pakistan)

Poverty Mapping

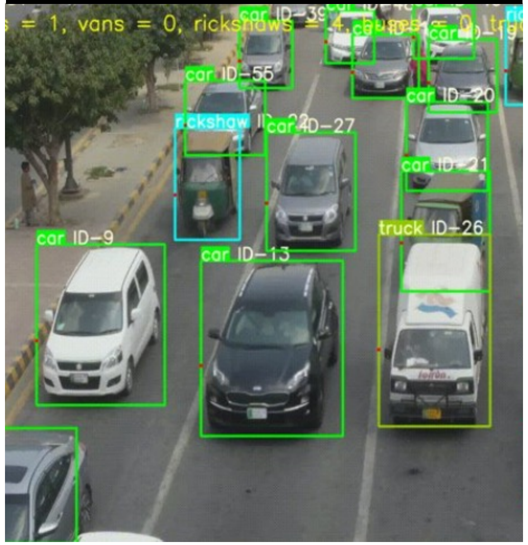


Bivariate map showing close correlation between nighttime lights and monthly income


Pakistan
India
Bangladesh

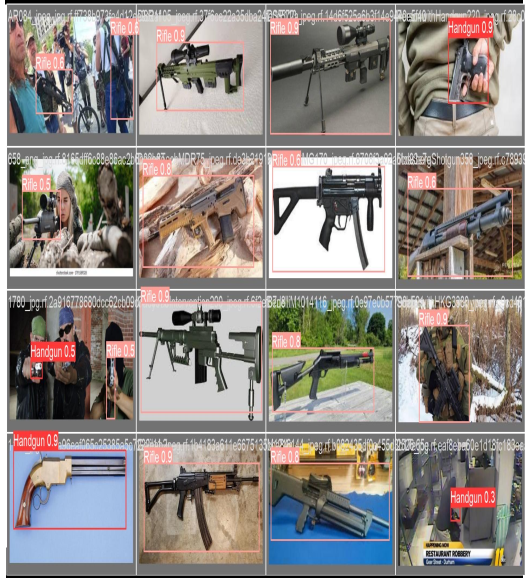
Vision-AI

Development of AI tools for traffic analytics



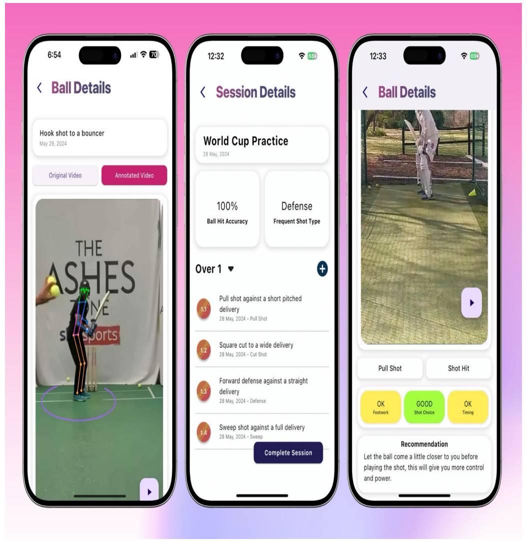
AI tool for traffic counting and classification

Firearms Detection



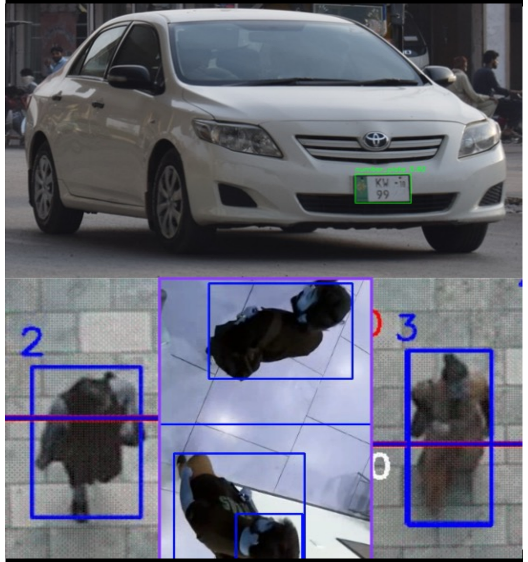
Automated Firearm Detection in Surveillance Footage Using Deep Learning

ShotSense: A cricket batting visualisation and training application



A Context-Aware Cricket Batting Recommendations System Using VGG16-Based Shot Classification and Gemini Large Language Model

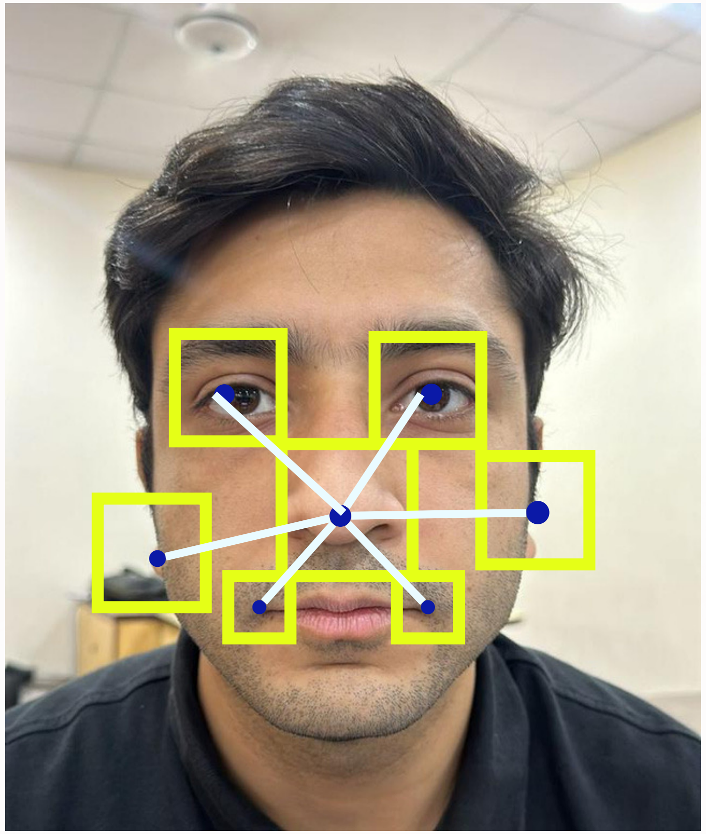
Mall Management System



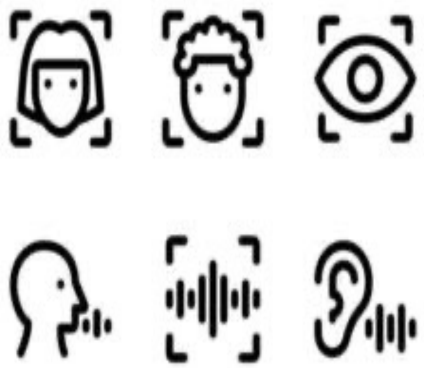
People counting and car plate tracking to enhance security and operational efficiency through real-time analytics.

Predict-AI

Personality Detection

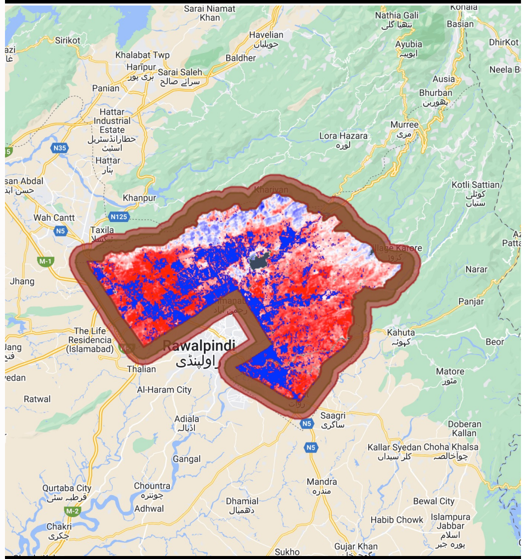


Personality Detection



PEER (Pre-Evaluated Efficient Resources) utilizes AI to evaluate face, handwriting, and voice.

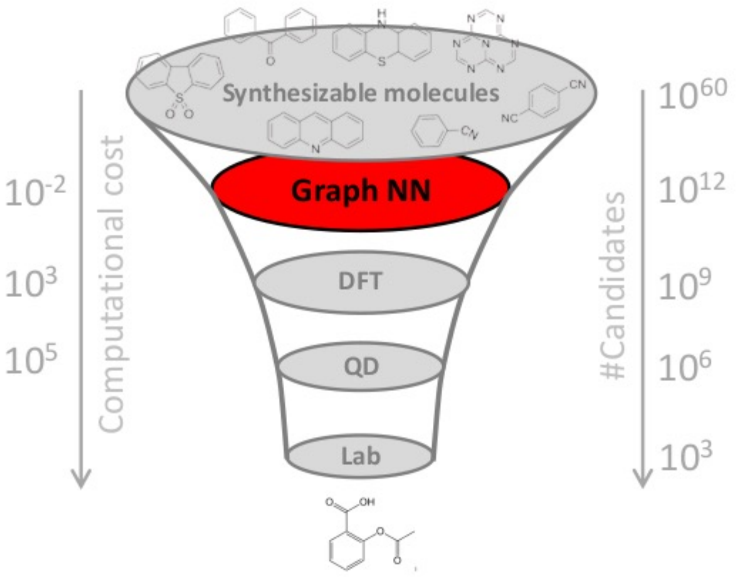
Surface Urban Heat Island (SUHI)



We can use LST to calculate a surface UHI (SUHI) intensity, including how it varies within cities at the pixel scale

Antibiotic Discovery

virtual drug screening



"Computational funnel"

Gilmer et al. 2017
Jin et al. 2020

What is Programming?

What is Programming?

Pro	**+**	**Gram**	**=**	**ProGram**
‘In Advance’	**+**	**‘Created’**	**=**	**Created in advance**

Traditional Programming


```c
int x = 10;
int y = 5;
printf("%d", x+y);
```

C code

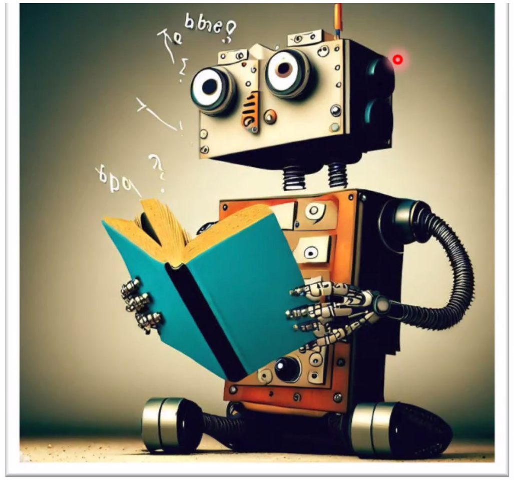
```python
x = 10
y = 5
print(x+y)
```

Python code

cpp Copy code

```cpp
#include <iostream>
using namespace std;

int main() {
    int num1, num2, sum;

    // Asking user to enter two numbers
    cout << "Enter first number: ";
    cin >> num1;

    cout << "Enter second number: ";
    cin >> num2;

    // Calculating the sum of two numbers
    sum = num1 + num2;

    // Displaying the result
    cout << "The sum of " << num1 << " and " << num2 << " is: " << sum << endl;

    return 0;
}
```

What is AI?

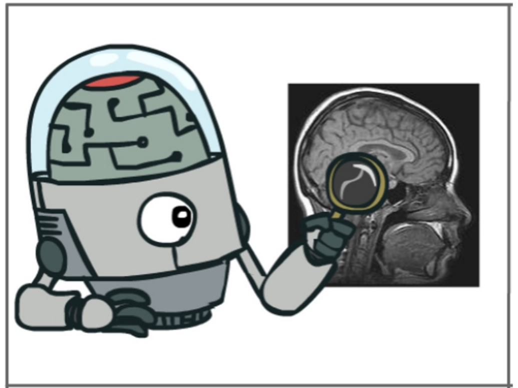


Machines that can learn from data and their own experience?

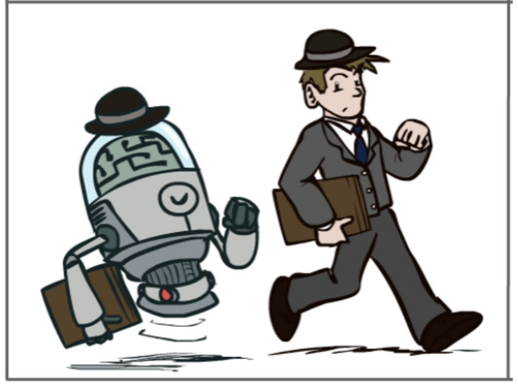
What is Artificial Intelligence (AI)?

The **science of making machines**/robots that:

Think like people



Act like people



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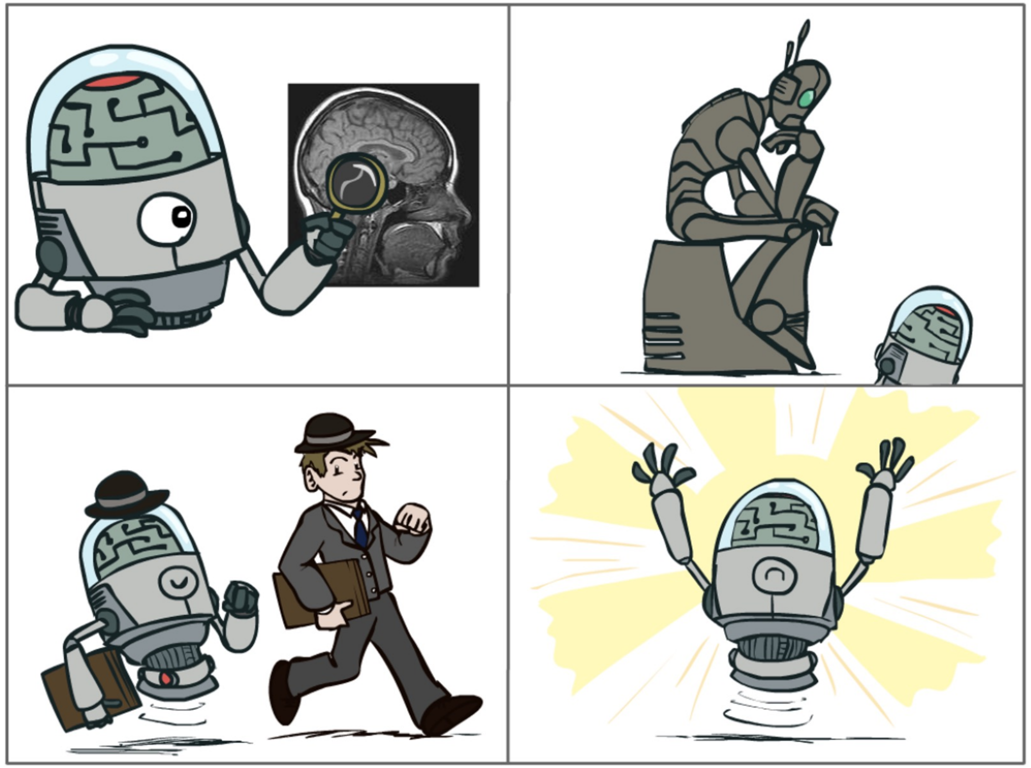
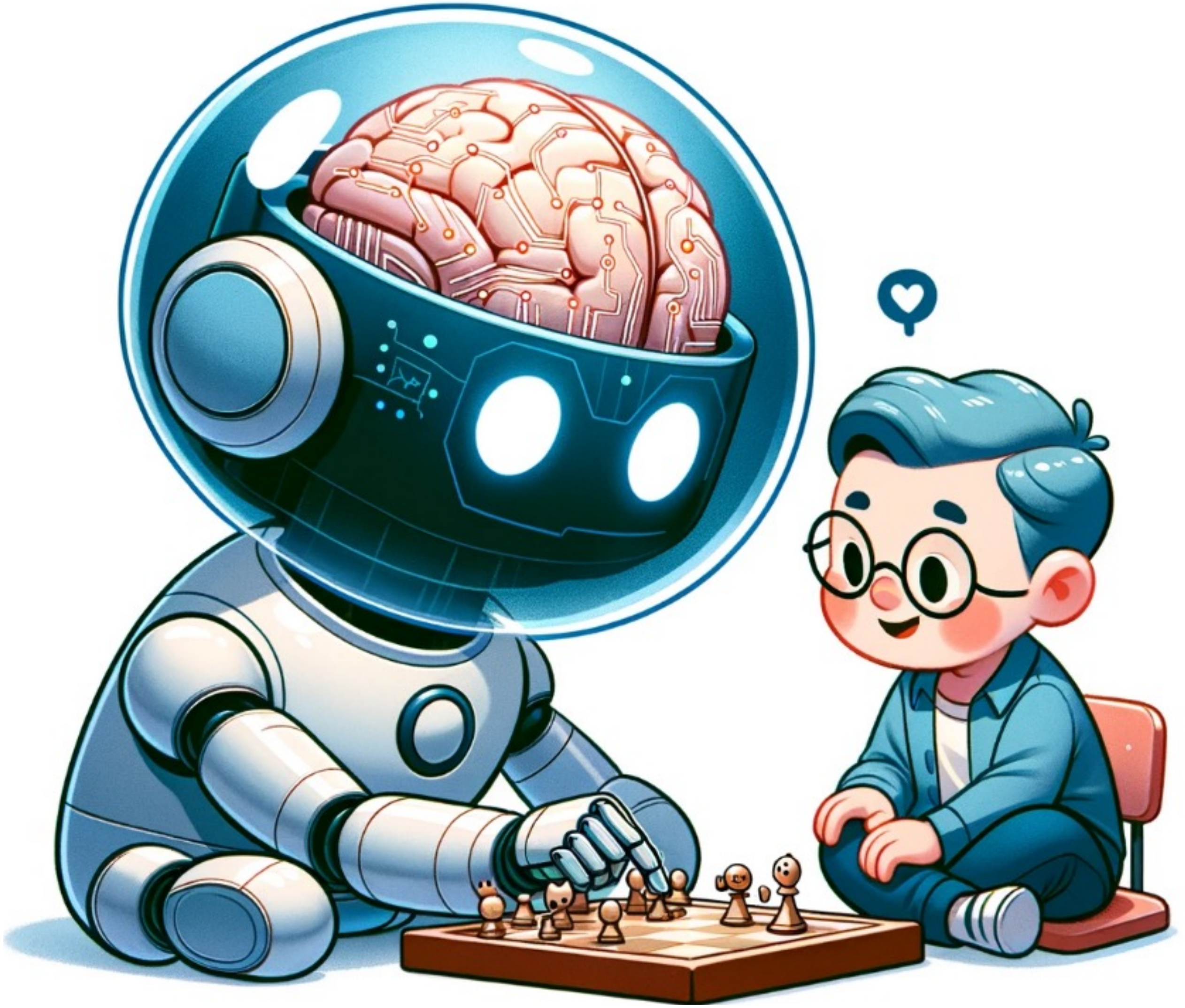


Think rationally

Act like people

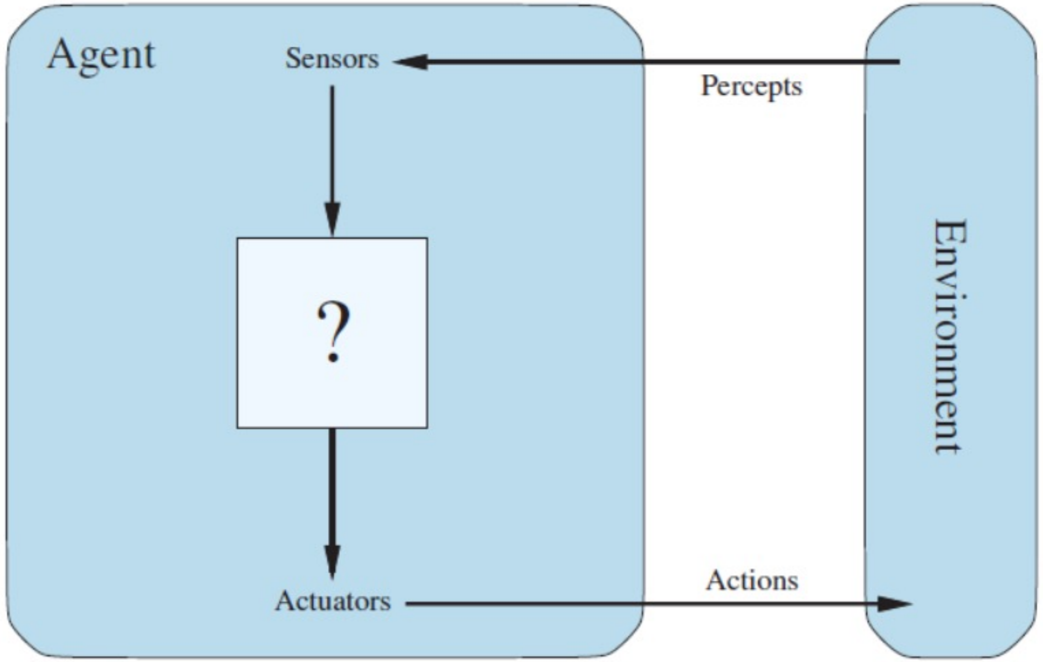
Act rationally

Rational: maximally achieving pre-defined goals



Agent

An agent is anything that can be viewed as perceiving its environment through **sensors** and acting upon that **environment** through **actuators**.

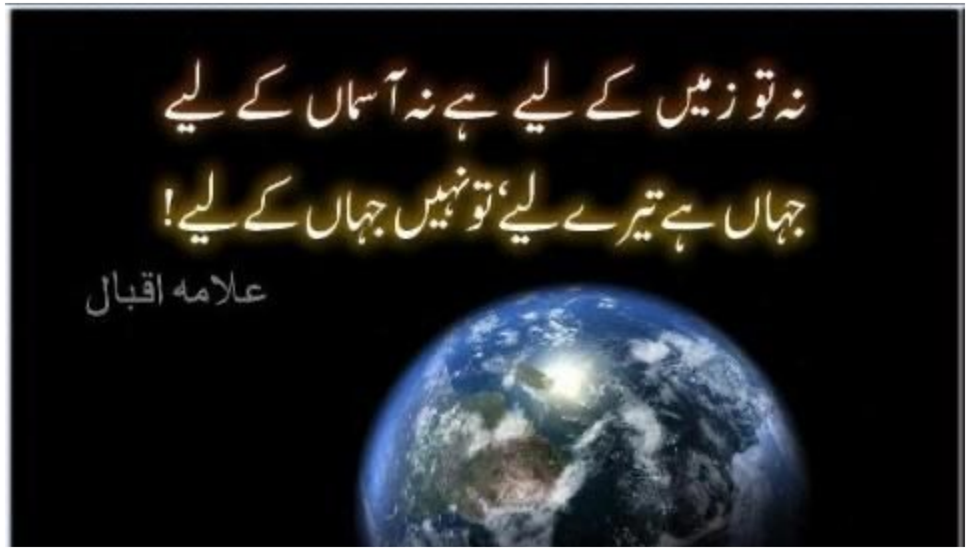


Human Agent

- Sensors:
- Actuators:
- Environment:

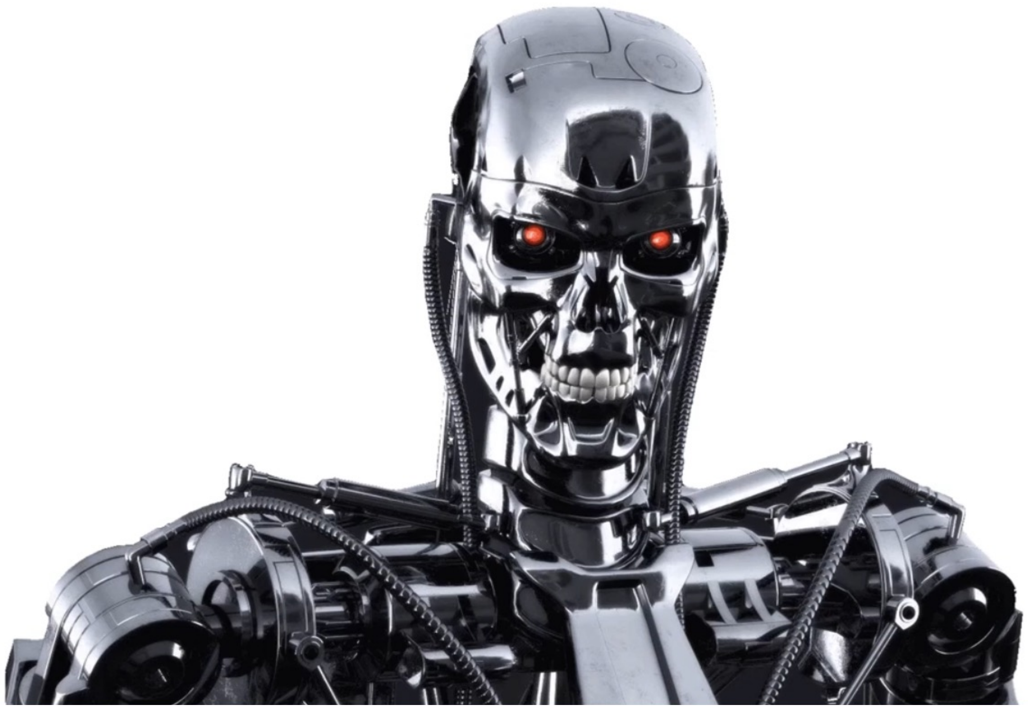
Human Agent

- Sensors: Eyes, nose, ears and other organs.
- Actuators: Hands, legs, vocal tract and so on.
- Environment: The entire universe



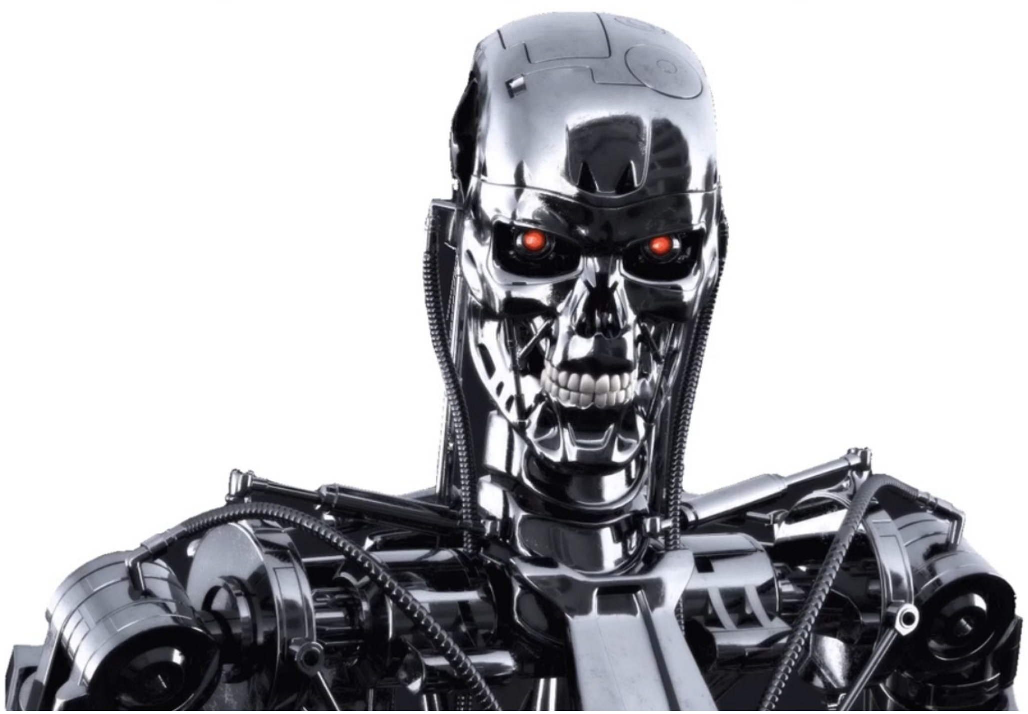
Robotic Agent

- Sensors:
- Actuators:
- Environment:



Robotic Agent

- Sensors: Cameras, infrared range finders, etc.
- Actuators: Various motors.
- Environment: The part of universe whose state we care about when designing this agent – the part that affects when the agent perceives and that is affected by the agent's actions.



Software Agent

- Sensors:
- Actuators:
- Environment:

Software Agent

- Sensors: Receives file contents, network packets and human input (keyboards, mouse, touchscreen, voice) as sensory inputs
- Actuators: Acts on the environment by writing files, sending network packets, and displaying information or generating sounds!
- Environment: The part of universe whose state we care about when designing this agent – the part that affects when the agent perceives and that is affected by the agent's actions.

What is Machine Learning (ML)?

Machine Learning is **function estimation** that fits well the training data.

What is Machine Learning (ML)?

Machine Learning is **function mapping between input (A) and output (B)**.

Size of house (Square feet)	Price (Lakhs)
523	115
645	150
708	210
1034	280
2290	355
2545	440

What is Machine Learning (ML)?

Input (A)	Ouput (B)	Application
Email	Spam? (0, 1)	?

What is Machine Learning (ML)?

Input (A)	Ouput (B)	Application
Email	Spam? (0, 1)	Spam Filtering

What is Machine Learning (ML)?

Input (A)	Ouput (B)	Application
Email	Spam? (0, 1)	Spam Filtering
Audio	Text transcript	?

What is Machine Learning (ML)?

Input (A)	Ouput (B)	Application
Email	Spam? (0, 1)	Spam Filtering
Audio	Text transcript	Speech Recognition

What is Machine Learning (ML)?

Input (A)	Ouput (B)	Application
Email	Spam? (0, 1)	Spam Filtering
Audio	Text transcript	Speech Recognition
English	Chinese	?

What is Machine Learning (ML)?

Input (A)	Ouput (B)	Application
Email	Spam? (0, 1)	Spam Filtering
Audio	Text transcript	Speech Recognition
English	Chinese	Machine Translation

What is Machine Learning (ML)?

Input (A)	Ouput (B)	Application
Email	Spam? (0, 1)	Spam Filtering
Audio	Text transcript	Speech Recognition
English	Chinese	Machine Translation
Ad, User info	Click? (0, 1)	?

What is Machine Learning (ML)?

Input (A)	Ouput (B)	Application
Email	Spam? (0, 1)	Spam Filtering
Audio	Text transcript	Speech Recognition
English	Chinese	Machine Translation
Ad, User info	Click? (0, 1)	Online Advertising

What is Machine Learning (ML)?

Input (A)	Ouput (B)	Application
Email	Spam? (0, 1)	Spam Filtering
Audio	Text transcript	Speech Recognition
English	Chinese	Machine Translation
Ad, User info	Click? (0, 1)	Online Advertising
Image, Radar info	Position of other cars	?

What is Machine Learning (ML)?

Input (A)	Ouput (B)	Application
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Audio	Text transcript	Speech Recognition
English	Chinese	Machine Translation
Ad, User info	Click? (0, 1)	Online Advertising
Image, Radar info	Position of other cars	Self-driving car

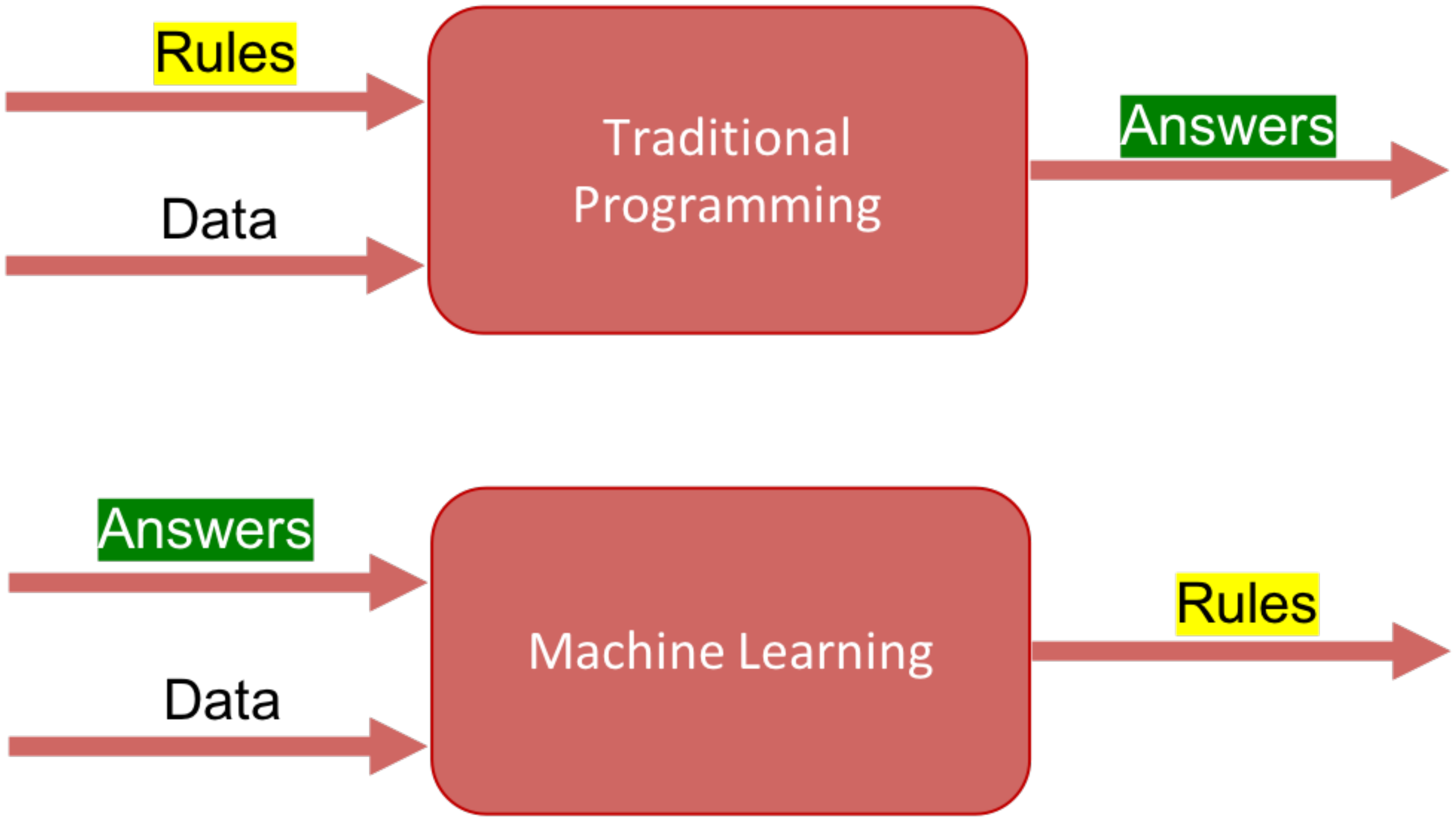
What is Machine Learning (ML)?

Input (A)	Ouput (B)	Application
Email	Spam? (0, 1)	Spam Filtering
Audio	Text transcript	Speech Recognition
English	Chinese	Machine Translation
Ad, User info	Click? (0, 1)	Online Advertising
Image, Radar info	Position of other cars	Self-driving car
Sequence of Words	Next Word	?

What is Machine Learning (ML)?

Input (A)	Ouput (B)	Application
Email	Spam? (0, 1)	Spam Filtering
Audio	Text transcript	Speech Recognition
English	Chinese	Machine Translation
Ad, User info	Click? (0, 1)	Online Advertising
Image, Radar info	Position of other cars	Self-driving car
Sequence of Words	Next Word	Chatbot

Traditional Programming vs AI



Questions

ai.bnu.edu.pk